1 Base Neuronal Dynamics

The presynaptic input signals decay as

$$\frac{dI_s}{dt} = -\frac{I_s}{\tau_s}, \qquad \frac{dx_{\text{pre}}}{dt} = -\frac{x_{\text{pre}}}{\tau_+}, \qquad \frac{dr_{\text{pre}}}{dt} = -\frac{r_{\text{pre}}}{\tau_r}.$$

These are fully determined by initial conditions set at the last presynaptic spike and enter the postsynaptic system as forcing functions. The postsynaptic dynamics are:

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\, w\, I_s(t), \tag{1}$$

$$\frac{dy_{\text{post}}}{dt} = -\frac{y_{\text{post}}}{\tau_-}, \tag{2}$$

$$\frac{dE}{dt} = -\frac{E}{\tau_e}, \tag{3}$$

$$\frac{dr_{\text{post}}}{dt} = -\frac{r_{\text{post}}}{\tau_r}, \tag{4}$$

$$\tau_{\bar{R}} \frac{d\bar{R}}{dt} = -\bar{R} - \big(r_{\text{post}} - \alpha\, r_{\text{pre}}\big)^2, \tag{5}$$

$$\frac{dw}{dt} = \Big[-\big(r_{\text{post}} - \alpha\, r_{\text{pre}}\big)^2 - \bar{R}\Big]\, E. \tag{6}$$

The postsynaptic state vector is $\mathbf{x}(t) = \big(V,\ y_{\text{post}},\ E,\ r_{\text{post}},\ \bar{R},\ w\big) \in \mathbb{R}^6$, and the inputs $\big(I_s(t),\ x_{\text{pre}}(t),\ r_{\text{pre}}(t)\big)$.

2 Pre-synaptic spike event

At each presynaptic spike time t^k_{pre} (an exogenous input event), the input subsystem variables receive discrete increments:

$$I_s \ \leftarrow\ I_s + 1, \qquad x_{\text{pre}} \ \leftarrow\ x_{\text{pre}} + 1, \qquad r_{\text{pre}} \ \leftarrow\ r_{\text{pre}} + 1.$$

The eligibility trace acquires an LTD contribution, weighted by the postsynaptic trace at the moment of the spike:

$$E \ \leftarrow\ E - \eta_-\, w\, y_{\text{post}}(t^{k-}_{\text{pre}}).$$

All remaining postsynaptic variables (V, y_{post}, r_{post}, $\bar{R}$, w) are continuous through t^k_{pre}; inter-spike dynamics (1)–(6) resume with the updated I_s.

3 Post-synaptic spike event

At each postsynaptic spike time t^k_{post}, defined by the threshold crossing $V = \theta$ with $dV/dt > 0$ (an endogenous event), the membrane potential resets and the postsynaptic trace variables receive discrete increments:

$$V \ \leftarrow\ V_{\text{reset}}, \qquad y_{\text{post}} \ \leftarrow\ y_{\text{post}} + 1, \qquad r_{\text{post}} \ \leftarrow\ r_{\text{post}} + 1.$$

The eligibility trace acquires an LTP contribution, weighted by the presynaptic trace at the moment of the spike:

$$E \;\leftarrow\; E + \eta_+ \,(w_{\max} - w)\, x_{\text{pre}}(t^{k-}_{\text{post}}).$$

All remaining variables ($\bar{R}$, w) are continuous through t^k_{post}. Following the spike, voltage integration is suspended for a refractory period τ_{ref}, after which it resumes from V_{reset}.

4 Complete Dynamics

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\, w\, I_s(t), \tag{7}$$

$$\tau_- \frac{dy_{\text{post}}}{dt} = -y_{\text{post}} + \rho_{\text{post}}(t), \tag{8}$$

$$\tau_e \frac{dE}{dt} = -E + \eta_+(w_{\max} - w)\, x_{\text{pre}}\, \rho_{\text{post}}(t) - \eta_-\, w\, y_{\text{post}}\, \rho_{\text{pre}}(t), \tag{9}$$

$$\tau_r \frac{dr_{\text{post}}}{dt} = -r_{\text{post}} + \rho_{\text{post}}(t), \tag{10}$$

$$\tau_{\bar{R}} \frac{d\bar{R}}{dt} = -\bar{R} - \big(r_{\text{post}} - \alpha\, r_{\text{pre}}\big)^2, \tag{11}$$

$$\frac{dw}{dt} = \Big[-\big(r_{\text{post}} - \alpha\, r_{\text{pre}}\big)^2 - \bar{R}\Big]\, E, \tag{12}$$

5 Timescale Separation and Relaxation Oscillations

The reward baseline $\bar{R}$ and synaptic weight w evolve on timescales much slower than the remaining state variables. The baseline has $\tau_{\bar{R}} = 5\,\text{s}$, which is $250\times$ slower than the membrane time constant $\tau_m = 20\,\text{ms}$. The weight w evolves according to $\dot{w} = M \cdot E$, where both the modulation signal $M = R - \bar{R}$ and the eligibility trace E (scaled by learning rates $\eta_\pm \sim 10^{-4}$) are small in magnitude, yielding an effective timescale for w that is comparable to or slower than $\tau_{\bar{R}}$.

5.1 Spike-count quantization

The postsynaptic neuron fires a discrete number of spikes $N(w)$ per presynaptic inter-spike interval $\Delta t_{\text{pre}} = 1/r_{\text{pre}}$. For a given weight w, the peak membrane drive is $R_m\, w\, I_s$, and the number of threshold crossings within each input cycle is a piecewise-constant, right-continuous function of w:

$$N(w) = \#\big\{t \in [t^k_{\text{pre}},\, t^{k+1}_{\text{pre}}) : V(t) \uparrow \theta\big\}.$$

At critical weights $w_c^{(n)}$ where N increments from n to $n+1$, the smoothed firing-rate trace r_{post} undergoes a step change of approximately $1/\tau_r$, and consequently the reward $R = -(r_{\text{post}} - \alpha\, r_{\text{pre}})^2$ experiences a discontinuous shift.

5.2 Mechanism of the limit cycle

Near a critical weight $w_c^{(n)}$, the system exhibits a relaxation oscillation with the following phases:

Slow recovery. With $w < w_c^{(n)}$, the neuron fires $N = n$ spikes per input cycle. The reward R and baseline $\bar{R}$ are approximately matched, so $M \approx 0$ and $\dot{w} \approx 0$. However, small residual $M > 0$ (since $\bar{R}$ lags R) causes w to drift upward on the timescale $\mathcal{O}(\tau_{\bar{R}}/(\eta\,|E|))$.

Threshold crossing. When w crosses $w_c^{(n)}$, an additional postsynaptic spike fires. The discrete increment $r_{\text{post}} \leftarrow r_{\text{post}} + 1$ causes R to jump sharply negative. Since $\bar{R}$ has not yet tracked this change ($\tau_{\bar{R}}$ is large), the modulation signal

$$M = R - \bar{R} = -\big(r_{\text{post}} - \alpha\, r_{\text{pre}}\big)^2 - \bar{R}$$

becomes strongly negative. Combined with a nonzero eligibility trace E, this drives $\dot{w} = M \cdot E \ll 0$, rapidly decreasing w.

Subcritical reset. With w pushed below $w_c^{(n)}$, the extra spike no longer fires and N returns to n. The reward R recovers, and $\bar{R}$ begins to track back toward the (less negative) steady-state R. Both $\bar{R}$ and w slowly recover, and the cycle repeats.

5.3 Non-convergence

The system cannot reach a stable fixed point at $w = w_c^{(n)}$ because the postsynaptic firing rate $r_{\text{post}}(w)$ is discontinuous in w at the critical values $w_c^{(n)}$, while the learning rule $\dot{w} = M \cdot E$ is continuous. The weight is attracted toward $w_c^{(n)}$ from below (where $M > 0$ drives w up) and repelled from above (where the extra spike makes $M \ll 0$ and drives w down). This constitutes a *grazing bifurcation*: the smooth flow interacts with a discontinuous reset map at the spike threshold, generically producing a stable limit cycle rather than a fixed point.

Equation Reference

Inter-spike dynamics — input subsystem:

$$\frac{dI_s}{dt} = -\frac{I_s}{\tau_s}, \tag{13}$$

$$\frac{dx_{\text{pre}}}{dt} = -\frac{x_{\text{pre}}}{\tau_+}, \tag{14}$$

$$\frac{dr_{\text{pre}}}{dt} = -\frac{r_{\text{pre}}}{\tau_r}. \tag{15}$$

Inter-spike dynamics — postsynaptic:

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m\, w\, I_s(t), \tag{16}$$

$$\frac{dy_{\text{post}}}{dt} = -\frac{y_{\text{post}}}{\tau_-}, \tag{17}$$

$$\frac{dE}{dt} = -\frac{E}{\tau_e}, \tag{18}$$

$$\frac{dr_{\text{post}}}{dt} = -\frac{r_{\text{post}}}{\tau_r}, \tag{19}$$

$$\tau_{\bar{R}} \frac{d\bar{R}}{dt} = -\bar{R} - \left(r_{\text{post}} - \alpha\, r_{\text{pre}}\right)^2, \tag{20}$$

$$\frac{dw}{dt} = \left[-\left(r_{\text{post}} - \alpha\, r_{\text{pre}}\right)^2 - \bar{R}\right] E. \tag{21}$$

Pre-synaptic spike event at t^k_{pre}:

$$I_s \;\leftarrow\; I_s + 1, \tag{22}$$

$$x_{\text{pre}} \;\leftarrow\; x_{\text{pre}} + 1, \tag{23}$$

$$r_{\text{pre}} \;\leftarrow\; r_{\text{pre}} + 1, \tag{24}$$

$$E \;\leftarrow\; E - \eta_-\, w\, y_{\text{post}}(t^{k-}_{\text{pre}}). \tag{25}$$

Post-synaptic spike event at t^k_{post}:

$$V \;\leftarrow\; V_{\text{reset}}, \tag{26}$$

$$y_{\text{post}} \;\leftarrow\; y_{\text{post}} + 1, \tag{27}$$

$$r_{\text{post}} \;\leftarrow\; r_{\text{post}} + 1, \tag{28}$$

$$E \;\leftarrow\; E + \eta_+\, (w_{\max} - w)\, x_{\text{pre}}(t^{k-}_{\text{post}}). \tag{29}$$

Model Parameters

Parameter	Description	Typical Value	Source
Membrane dynamics (leaky integrate-and-fire)			
τ_m	Membrane time constant	20 ms	Dayan and Abbott, 2001; Koch and Segev, 1998; McCormick et al., 1985
E_L	Leak (resting) potential	−70 mV	Destexhe et al., 2001; McCormick et al., 1985
R_m	Membrane resistance	100 MΩ	Dayan and Abbott, 2001; Koch and Segev, 1998
θ	Spike threshold	−55 mV	Destexhe et al., 2001; Henze et al., 2000
V_{reset}	Reset potential after spike	−65 mV	Destexhe et al., 2001; Gerstner et al., 2014
τ_{ref}	Absolute refractory period	2 ms	Gerstner et al., 2014; Henze et al., 2000
Synaptic current			
τ_s	Synaptic current decay (AMPA)	5 ms	Dayan and Abbott, 2001; Destexhe et al., 1994; Roth and van Rossum, 2009
STDP trace time constants			
τ_+	Pre-synaptic LTP trace	20 ms	Bi and Poo, 1998; Song et al., 2000
τ_-	Post-synaptic LTD trace	20 ms	Bi and Poo, 1998; Song et al., 2000
Firing-rate trace			
τ_r	Low-pass firing-rate trace	200 ms	Gerstner et al., 2014; Izhikevich, 2006
Eligibility trace			
τ_e	Eligibility trace decay	1 s	Frémaux and Gerstner, 2016; Gerstner et al., 2018
Reward baseline			
$\tau_{\bar{R}}$	Running reward average	1–10 s	Frémaux and Gerstner, 2016; Gerstner et al., 2018; Sutton and Barto, 1998
α	Target firing-rate ratio	0.5	—
Plasticity learning rates			
η_+	LTP learning rate	0.01	Morrison et al., 2008; Song et al., 2000
η_-	LTD learning rate	0.0105	Morrison et al., 2008; Song et al., 2000
Synaptic weight bounds			
$w_{\max}$	Maximum synaptic weight	1.0 (normalized)	Morrison et al., 2008; Song et al., 2000
$w(0)$	Initial synaptic weight	$0.5\, w_{\max}$	Abbott et al., 2016; Song et al., 2000